

Execution of Modal Triplet Theory II: Flavor, CKM, Neutral, and Higgs Status after True-SM Closure

Peter Nero

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Abstract

We replace the legacy fitted Calabi–Yau flavor benchmark with the current audited execution status. The 27-by-27 qutrit–Weyl/minimal matrix ledger, charged Yukawa profile rows, selected Π_{CKM} rows, electroweak primitive, and direct Higgs/threshold row support embedded renormalized-Standard-Model equivalence at the adopted one-shared-physical-primitive/profile standard. The final audit closes 12/12 obligations at that standard. This is not a zero-knob derivation of measured masses. The nine charged Yukawa magnitudes and λ_H remain profile labels rather than internally emitted strict source values. CKM closes at the prediction-with-uncertainty standard, with maximum displacement 2.3565×10^{-4} standard deviations. In the neutral sector, MTT now emits a complete internal dimensionless response, but its spectrum [1, 4, 7] cannot be converted into the physical neutrino spectrum by one common scale or nil subtraction. The old fitted PMNS, seesaw, and few-TeV Higgs benchmarks are therefore withdrawn as predictions.

Revision note for this edition

Supersedes. *Execution of Modal Triplet Theory II: Flavor, CKM/PMNS, and Higgs Sector on the CY Corner*, version 2.

Reason. Fitted Calabi–Yau flavor matrices, seesaw assumptions, and a few-TeV Higgs benchmark were reported too strongly and no longer represented the current finite-operator calculation.

Resolution. Version 3 records the locked 27×27 , charged-Yukawa, CKM, electroweak, and Higgs/threshold profile packets, and proves the present one-scale/nil-subtraction obstruction in the neutral sector.

Retained result. Flavor and Higgs data close embedded renormalized-SM equivalence at the declared one-shared-primitive/profile standard.

Remaining boundary. Strict source emission of profile magnitudes and absolute neutral scale/ontology remains beyond that standard.

1 Closure standard

The accepted result is embedded renormalized-SM equivalence on the selected branch at the one-shared-physical-primitive/profile standard. It imports the standard SM quantization and uses a declared measured-input profile. It does not derive all empirical values from MTT, prove unique observed-branch selection, or establish a zero-primitive/no-knob theory.

The older Execution II matrices were constructed with adjustable local texture entries, phases, scale choices, and benchmark masses. The first correction pass made their real-matrix

diagonalizations arithmetically reproducible and removed an unsupported CP claim. The present revision goes further: those matrices are classified as historical profile fits and are no longer the authority for flavor closure.

2 Charged flavor and the 27-by-27 operator

The current charged-sector authority contains:

- the locked 27-by-27 putrit–Weyl/minimal matrix ledger;
- the counted AH-equivalent HYM/projective lane, closed 8/8;
- the selected charged Yukawa basis and magnitude-profile rows; and
- the selected common-scheme threshold and mass transport with covariance.

These objects establish that the selected branch can reproduce the charged renormalized-SM data coherently through one operator architecture.

At the strict source-value tier, however, the magnitude-bearing functional has ten replay labels—nine charged Yukawa magnitudes and λ_H —and zero accepted internal no-knob labels. Thus the matrix and basis maps are genuine structural achievements, while the measured charged masses remain profile data at the adopted closure standard.

3 CKM and weak CP violation

Three selected Π_{CKM} source rows are retained. They use neither observed CKM values nor target residual fitting as selectors. Against the frozen declared CKM uncertainty profile, their maximum displacement is

$$2.3564680386 \times 10^{-4} \text{ standard deviations.} \quad (3.1)$$

The strict-upgrade row U4 therefore closes at the prediction-with-uncertainty-profile standard. Exact equality to a moving global fit central estimator is neither obtained nor required, and no empirical residual correction is introduced.

This CKM result supersedes the legacy printed real Yukawa matrices. Those real matrices could reproduce mixing magnitudes but could not establish a nonzero Jarlskog invariant. Weak-sector CP now belongs to the selected complex branch and Π_{CKM} execution. It must not be confused with strong-CP closure, which remains open.

4 Neutral and PMNS sector

The selected same-source neutral construction has advanced beyond the old benchmark seesaw. Its complete internal dimensionless response is

$$a_{\text{int}} = 0.34195899479289005, \quad dY_\nu = I_3 + X_3, \quad (4.1)$$

with

$$H_{1,\nu} = \begin{pmatrix} -2a & 0 & -2a \\ 0 & -2a & -2a \\ -2a & -2a & 0 \end{pmatrix}. \quad (4.2)$$

All nine $H_{1,\nu}$ rows and all seven provenance fields are selected: six entries equal $-2a$ and three are exact zeros. The associated selected two-representative orbit has diagonal coefficient 1, cyclic-shift coefficients $3/2 \pm i\sqrt{3}/2$, and Hermitian spectrum $[1, 4, 7]$.

This is not yet a dimensionful neutrino Yukawa or mass matrix. A common scale cannot alter eigenvalue ratios. Nil subtraction gives $[0, 3, 6]$ and therefore the invariant ratio

$$r_{\text{direct}} = \frac{3}{6} = \frac{1}{2},$$

whereas the stored normal-ordering oscillation postcheck is $r_{\text{post}} = 0.02980501393$. The scale-only completion is therefore falsified. The minimal surviving routes require either a selected non-affine spectral-action slope plus one universal physical scale, or a selected dimensionful seesaw/Dirac–Majorana block.

Consequently the old fitted neutrino masses, PMNS matrix, and Majorana scale are withdrawn as predictions. They may be used only as historical benchmark data; they do not close absolute neutrino mass, ordering selection, or ontology.

5 Higgs status

At the adopted profile standard, P_{EW} is counted once as the one shared physical primitive, with zero Higgs-specific free parameters. The direct $K_{\text{threshold}} \cdot \Omega_H \cdot \lambda$ row and its row-purpose/formula bridge are locked, and the common SMDR transport carries the Higgs row in the selected precision object.

This closes the Higgs coordinate for embedded renormalized-SM equivalence. It does not make λ_H an internally emitted no-knob value: the magnitude/profile functional still carries it as a replay label. The former $\tan \beta = 10, 5$ TeV supersymmetric matching benchmark and its one-loop Higgs pole-mass corridor are withdrawn. A strict UV-Higgs derivation would require a selected magnitude-bearing source functional and dedicated scheme/threshold transport independent of the measured Higgs coordinate.

6 Sector status table

Sector	Closed result	Remaining stronger obligation
Charged flavor	27-by-27 structure, basis maps, profile magnitudes	internal source emission of nine magnitudes
CKM	selected prediction profile; U4 closed	no stronger central-estimator identity required
Neutral/PMNS	internal dimensionless 9/9 response	non-affine action or dimensionful seesaw, scale and ontology
Higgs	direct K row and profile-standard coordinate	internal no-knob λ_H value source

7 Conclusion

Execution II now records the actual frontier. Charged flavor, CKM, and Higgs participate in the closed 12/12 embedded renormalized-SM equivalence theorem at the adopted one-shared-primitive/profile standard. The CKM strict upgrade is also closed at the correct uncertainty-profile criterion.

The result is not a derivation of all fermion masses from zero empirical inputs. Ten magnitude labels remain profile data at the strict source tier, and physical neutrino completion is demonstrably impossible by a common scale alone. This separation preserves the genuine matrix/operator achievements without promoting fitted or replayed values into source theorems.

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